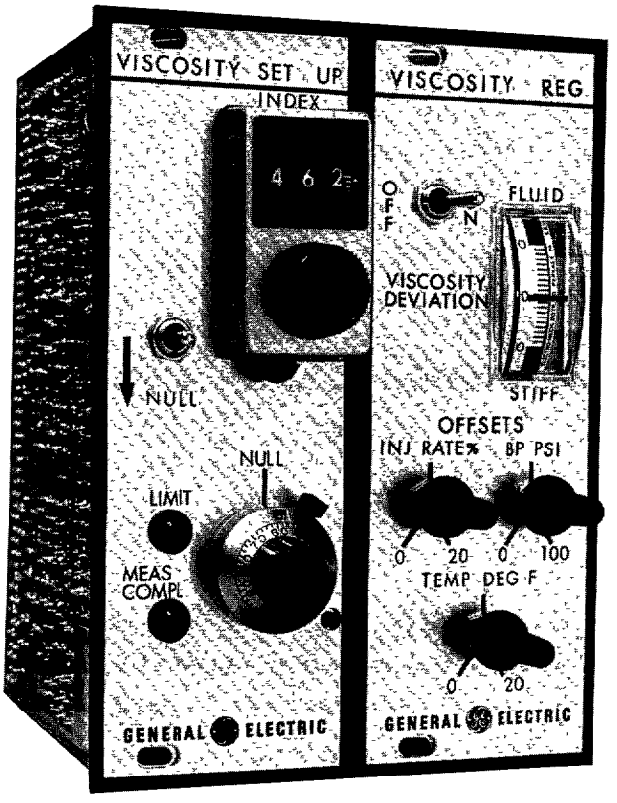
GEK-40680

INSTRUCTIONS PM1000* CONTROL

MELT VISCOSITY REGULATION FOR PLASTIC PROCESSING MACHINES TYPE IC7966



The information contained herein is intended to assist users and dealers in the servicing of control furnished by the General Electric Company. It does not purport to cover all details or variations in equipment or provide for every possible contingency to be met in connection with installation, operation or maintenance.

Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purpose, the matter should be referred to the machinery manufacturer through his normal service channels.

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PM1000* CONTROL

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SECTION 1 INTRODUCTION

1.1 PROCESS CONTROL

Process Control is used to refer to a group of functions that is aimed at making the process more consistent and more controllable. Some process control functions are related to increased control of the machine. Examples are Programmed Injection and Hydraulic Pressure or Flow Control. Other process control functions relate to a condition in the process and provide a signal or an adjustment to the machine controls. Examples are Cavity Event Detection, Melt Viscosity Control and Cavity Pressure Regulation.

1.2 SEQUENCE CONTROL

Process Control adds to the basic controls of time, temperature and sequence. All functions in this book are described as being connected to a PM1000 sequence control.

1.3 POWER SUPPLY

The process control functions are designed to use the same power supply as the sequence control and, where loading permits, actually share the sequence control power supply.

1.4 SEQUENCE CONTROL INTERFACE

The PM1000 process control logic is compatible with the PM1000 Flo-Thru sequence control logic. Hence, wiring is the only required interface between the two controls.

1.5 METERLESS TROUBLESHOOTING

One of the successes of the PM1000 Flo-Thru control is the logic indicating lights that permit troubleshooting without using a meter. This same approach is carried into the PM1000 process control with indicating lights and card front meters. The card front meters also permit installation adjustments to be made without an external meter.

1.6 OFF-ON SWITCHES

Each process control function has its individual "off-on" switch, rather than a master process control "off-on" switch. This provides complete flexibility over the combinations of functions used. When a function is in the "off" position, the indicating lights are permitted to operate as normal for set-up reasons; however, output correction or alarm signals are disabled.

1.7 VALVES AND TRANSDUCERS

Process control functions can require such external hardware as electrically controllable valves for pressure or flow, hydraulic pressure transducers, or cavity pressure transducers. If so, the required hardware is described in the appendices.

1.8 LOW-LEVEL WIRING

The electrical connections to the valves and transducers mentioned above are low-voltage wires. They must be shielded as indicated by the elementary diagram and kept separated from higher voltage wiring.

1.9 CAUTION FOR REMOVING OR REINSERTING CARDS

It is strongly recommended that power be removed from the panel prior to either removing or reinserting printed-circuit cards. Removing or inserting a card without removing power can damage the card due to transient conditions.

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INTRODUCTION (Continued)

1.10 GENERAL MAINTENANCE

These instructions are detailed in the standard sequence manual GEK-40400. They should be reviewed in conjunction with using this process control instruction book.

1.11 OTHER PROCESS CONTROL MANUALS

Available PM1000 process control instruction books are listed in Appendices 1. The books included in the PM1000 panel should match the functions in that panel.

SECTION 2

MELT VISCOSITY REGULATION

2.1 WHAT VISCOSITY REGULATION DOES

Material viscosity, long recognized as a key process parameter in injection molding, is consistently maintained at an optimum index number as determined in initial machine set-up. Viscosity in plastics material is inversely proportional to temperature, back pressure, and injection rate.

2.2 GENERAL OPERATION

The Melt Viscosity Control measures the viscosity of the molten plastic during injection. If this measured value has changed from a preset value, which had been determined as making good quality parts, the melt viscosity control will automatically adjust the injection rate, the back pressure, and the barrel temperatures to return the viscosity to the preset value.

2.3 MACHINE DEVICES REQUIRED

Melt Viscosity Control requires an electrically controlled flow valve for controlling the injection rate, an electrically controlled pressure valve for controlling the back pressure, a hydraulic pressure transducer for measuring the injection pressure, and a potentiometer connected to the injection cylinder to indicate position. This position signal is converted into a speed measurement by the control. A description of how these devices operate and the compatible models are described in the appendices.

2.4 FEATURES REQUIRED IN SEQUENCE CONTROL

The sequence control must contain provisions to end the injection boost period by some means in addition to the standard boost timer. This can be achieved by a screw-forward limit switch or by a process control function called Event Detection. This provision is necessary to permit the viscosity control to change the injection rate without causing short shots or flash, because when changing the injection rate, time cannot be used to control the size of the shot. For either the limit switch or the event detector, the preferred operation is for the boost timer to be set a few seconds longer than normal. It then serves as a monitor to limit the boost time in the event of an injection malfunction such as a clogged nozzle.

Viscosity Control requires that the sequence control contain a solid-state limit switch for the screw-back position. This requirement is further explained in the Description of Operation section.

2.5 OTHER PROCESS CONTROL FUNCTIONS REQUIRED

Melt Viscosity Control requires two other process control functions: Hydraulic Pressure Control and Hydraulic Flow Control. The functions are required to permit the viscosity control to automatically change the injection rate and back pressure.

2.6 FUNCTIONAL DIAGRAM

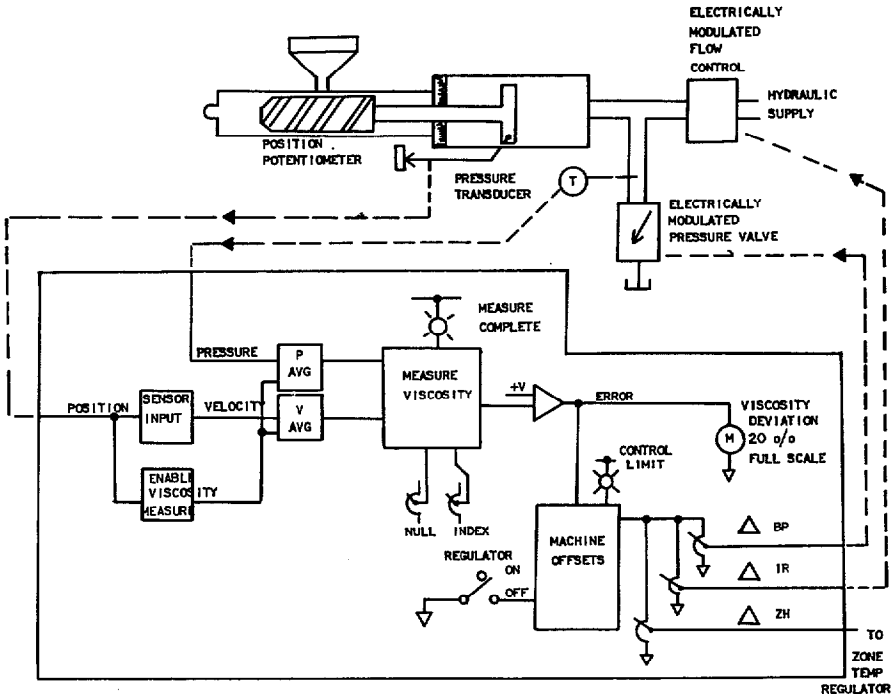


Fig. 1. Functional Diagram for Melt Viscosity Regulation

SECTION 3
DESCRIPTION OF VISCOSITY CONTROL CARDS

3.1 CATALOG NUMBERS

The circuitry for Melt Viscosity Control is contained on three printed circuit cards:

IC3622GVSA — Viscosity Set-Up

IC3622GVRA — Viscosity Regulator

IC3622GVRB — Viscosity Initiate

NOTE: *The IC3622GVRB is an auxiliary plug-in card that mounts under the GVRA card.*

3.2 DESCRIPTION OF GVSA CARD (Viscosity Set-up Card)

This card provides most of the operator set-up devices for the function. It receives averaged hydraulic pressure and ram velocity signals from the GVRA card and uses them to determine the viscosity measurement. It also contains the circuitry for comparing this measurement to the index value and generating a correction signal.



IC3622GVSA *IC3622GVRA*

DESCRIPTION OF VISCOSITY CONTROL CARDS (Continued)

3.3 GENERAL DESCRIPTION OF GVRA CARD (Viscosity Regulator Card)

This card averages the hydraulic pressure signal and the ram velocity signal for a determined sample time during the injection forward motion. It also provides a meter to display the viscosity deviation from set-point and single-turn pots to control the maximum regulator offsets to temperature, back pressure, and injection rate. It also contains an "off/on" switch to disable the viscosity regulator during machine set-up.

3.4 GENERAL DESCRIPTION OF GVRB CARD (Viscosity Initiate Card)

This card provides the signal that initiates the viscosity measurement. It does this by looking at the setting of the screw-back static limit switch and the maximum screw-forward position. This card provides a signal when the screw has traveled 25 percent of this distance from the screw-back position. Although the screw-forward position can be a static limit-switch setting, generally the signal wired to this input is the -8.2-volt signal that represents the forward mechanical limit. (Normal application is -8.2 volts = full forward = 000 percent).

SECTION 4

DESCRIPTION OF OPERATION

4.1 TYPICAL ELEMENTARY SHEET

See Figure 2.

4.2 INPUT AND OUTPUT SIGNALS

These signals are described in Figure 2.

4.3 NORMAL SEQUENCE

At the start of injection, the measurement-complete light goes off (pin 35 on GVRA goes to logic 1) and the average pressure (pin 20 on GVRA) and average velocity outputs (pin 28 on GVRA) reset to 0 volts. The Viscosity Deviation meter will deflect off scale at this time.

When the screw has traveled forward 25 percent of the distance from the screw-back point, the GVRB card provides a "Start Viscosity Measure" signal (logic 0 on pin 41 of GVRB). This signal starts a time delay on the GVRA card which determines the end of the viscosity measurement. This time delay is 300 milliseconds. Optional connections can change this to either 100 or 1000 milliseconds. At the end of the viscosity measurement period, the value of the viscosity measurement is determined from the stored average pressure and velocity signals. The Viscosity Deviation meter displays the deviation of the measurement from the setpoint value. The meter is normally connected for full scale = 20 percent deviation. Alternate connections for 10 percent and 5 percent can be made.

Since the average pressure and average velocity signals are held by analog storage circuits, the Viscosity Deviation meter maintains its display until the start of the next injection cycle.

The viscosity deviation signal feeds into an analog stepping circuit that generates a correction signal (pin 41 on GVSA) that can automatically alter three machine set-up functions, i.e. injection rate, back pressure and zone temperatures. The analog stepping circuit stores this correction signal. At the end of the next measurement, this correction signal is increased or decreased as required, by an amount proportional to the magnitude of the new error. Therefore, the control makes a large correction for a large error and a small correction for a small error.

As the screw retracts, the start viscosity measurement signal returns to a logic 1 in preparation for the next injection cycle.

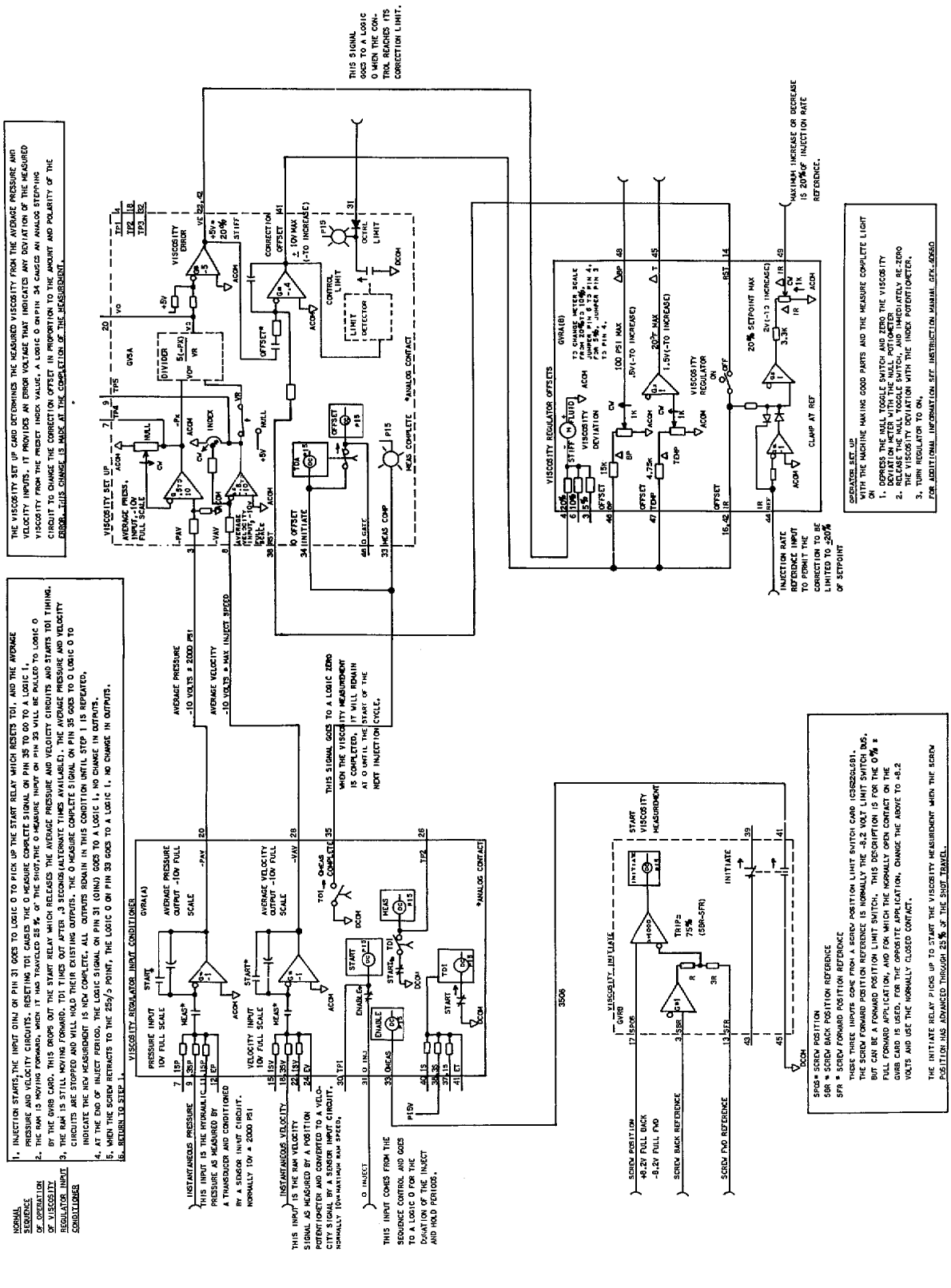


Fig. 2

DESCRIPTION OF OPERATION (Continued)

4.4 USE ON COLD MACHINE START

Once the machine and Viscosity Regulator have previously been operating on a given part set-up, the Viscosity Regulator will help the machine make parts more quickly on a cold start. The Viscosity Regulator will probably go into limit during this time due to the large error. Do not reset the limit at this time, because as the machine warms up, the regulator will make corrections to reduce the error and will back itself out of the limit.

4.5 FAULT SEQUENCE

If an error has not been corrected after several injection cycles, the Control Limit light will come on to indicate that the control has applied its total correction. This could mean that a trend has set up in the process and the control has been forced to consistently make corrections in the same direction. In this case, check the machine set up and re-set up the Viscosity Regulator. Losing a heater band or a hot runner heater can cause this type of behavior, or the fault light can mean that the control is not correcting. (See Troubleshooting Section for additional information).

SECTION 5
INSTALLATION ADJUSTMENTS

There are no installation adjustments to the Viscosity Regulator function. The Closed Loop Flow and Pressure functions, however, must be properly adjusted before operating the Viscosity Regulator.

SECTION 6
OPERATOR SET-UP

6.1 GENERAL INFORMATION

Before the Viscosity Regulator can be set-up for a given part, the machine must be cycling with good parts being produced. This implies that the machine has been set-up using normal values of injection rate, back pressure and zone temperatures. For the Viscosity Regulator to perform at its optimum, the injection pressure must be set at a high enough value that it does not restrict the Injection Rate flow valve.

Generally there is a second set-up requirement for the Viscosity Regulator to perform at its optimum. The switch-over signal from Inject Pressure to Hold Pressure must be controlled either by a boost override limit switch or by a Cavity Event Detector. This is necessary to permit the Viscosity Regulator to change the Injection Rate without causing short shots or flashing.

For either the boost override limit switch or the Cavity Event Detector, a good set-up is for the Boost Timer to time out 3 to 5 seconds after the switch-over normally occurs. The Boost Timer then serves as a back-up in the event the limit switch is not reached, or the cavity pressure does not reach the desired level.

6.2 VISCOSITY REGULATOR SET-UP

Turn the off/on toggle on the Viscosity Regulator to "off." With the machine making good parts and while the Measure Complete light is on, set-up the Viscosity Regulator by depressing the Viscosity Null toggle switch and zeroing the Viscosity Deviation meter by operating the Null Potentiometer. Releasing the Null toggle switch should then cause the viscosity deviation to move away from zero.

Re-zero the Viscosity Deviation meter at that time, utilizing the Viscosity Index reference. This should be done on the same shot. However, it must be completed while the Measurement Complete light is on. It would be well to recheck the null and Index settings once more. Unless the material being used is extremely heat-sensitive, set the Injection Rate, Back Pressure, and Temperature offset adjustments at mid-range, which will then allow the control a correction range of 10 percent of setpoint on Injection Rate, 50 psi on Back Pressure, and 10 degrees F on the barrel regulators in order to regulate the melt viscosity. At this point the off/on toggle on the Viscosity Regulator can be operated to the "on" position, and the Viscosity Regulator will now regulate the viscosity to this set-up condition.

OPERATOR SET-UP (Continued)

It should be noted that for best operation of this function, the basic machine set up on the Injection Rate should not be less than 20.0 percent or greater than 80.0 percent. Likewise, the Back Pressure should not be set at its minimum value.

If the machine set-up is manually changed, it may be necessary to turn the Viscosity Regulator to "off" to clear it, let the machine stabilize, set up the Viscosity Regulator to the new condition, and turn it back to "on."

If power is removed and re-applied with the Viscosity Regulator toggle in the "on" position, it is desirable to operate the switch to the "off" position momentarily to clear the analog storage circuit. However, if this is not done, the regulator will do it automatically over several shots.

SECTION 7
AVAILABLE FUNCTIONAL MODIFICATIONS

7.1 VISCOSITY MEASURING TIME

The viscosity measuring time can be changed from 300 milliseconds to 100 milliseconds by moving the wires from pin 9 to 11, pin 18 to 22, and pin 38 to 37 on the Viscosity Regulator (GVRA) receptacle. This change might be required on a machine with a shot time less than 1 second to insure that the measurement is complete before the injection motion stops.

The viscosity measuring time can be changed from 300 milliseconds to 1000 milliseconds by moving the wires from pin 9 to 7, pin 18 to 15, and pin 38 to 40. This change might be made on a large machine having a shot time greater than 6 or 7 seconds to permit additional averaging of the measurement.

The viscosity control cannot operate properly if the injection time is less than 1.3 times the viscosity measuring time. If this condition arises, either the wiring must be changed to provide a shorter measuring time, or the injection rate slowed down sufficiently. This problem is most likely to arise when only a very small percentage of the barrel stroke is required to fill the mold.

7.2 VISCOSITY DEVIATION SENSITIVITY

The full-scale deviation can be changed from 20 percent to 10 percent by adding a wire from pin 4 to 6 on the Viscosity Regulator (GVRA), or to 5 percent by adding a wire from pin 4 to pin 3. The increased sensitivity might be desirable for an extremely critical part.

SECTION 8
TROUBLESHOOTING

8.1 POWER SUPPLY CONNECTIONS

A. Viscosity Regulator Card IC3622GVRA

1. P15V on pin 27
2. N15V on pin 29
3. DCOM on pin 1
4. ACOM on pin 2

B. Viscosity Initiate Card IC3622GVRB

5. P15V on pin 27
6. N15V on pin 29
7. DCOM on pin 1

C. Viscosity Set-Up Card IC3622GVSA

8. P15V on pin 27
9. N15V on pin 29
10. DCOM on pin 1
11. ACOM on pin 2

TABLE 8.2
TROUBLESHOOTING CHART

SYMPTOMS	PROBABLE CAUSE	REMEDY
VISCOSITY DEVIATION Will Not Null Per the Set-up Instructions	Missing or loose power supply connection on one of the three viscosity cards.	Verify per 8.1 of this section.
	Defective Viscosity Regulator, Set-up, or Initiate card.	Replace card.
	The Viscosity initiate signal is missing.	Verify that Pin 33 on the Viscosity Regulator is being switched to zero as the ram has traveled forward 25 percent of the shot.
	The injection time is less than 1.3 times the viscosity measuring time.	Slow down the injection rate sufficiently, or see page 7.1 on wiring change to modify viscosity measuring time.
VISCOSITY DEVIATION Swings Widely From Shot to Shot	Flow Control Function is not working properly or Flow Hydraulic Regulator is turned off.	Turn on and verify operation. Review instructions on Closed Loop Flow Control if necessary.
	Verify common wire on Pin 2 of the Viscosity Set-up and Viscosity Regulator cards.	Reterminate if loose or missing.
VISCOSITY MEASUREMENT No Change From Shot to Shot	Measurement is not being initiated.	Verify that the Measure Complete light goes out at the start of each injection shot.
	The machine is extremely stable, and the viscosity is not changing.	Increase the viscosity deviation sensitivity (See Section VII).
	The hydraulic pressure signal is missing.	Verify by observing the meter indication on the sensor input card. (Toggle must be in "PRES" position).
	The ram velocity signal is missing.	Verify by observing the meter indication on the sensor input card. (Toggle must be in "VEL" position).
		Verify the presence of a 0 to -10 V average velocity signal on pin 20 of the Viscosity Regulator Card. If not present replace that card.
VISCOSITY REGULATOR Does Not Correct	Δ IR, Δ BP and Δ ZH RANGE POTS on the Viscosity Regulator are set full CCW.	Turn CW sufficiently to permit corrections.
	Δ ZH signal does not offset the temperature regulators. Verify this by rotating the Δ ZH Range pot on the Viscosity Regulator and confirming that the deviation meters on the temperature regulators do not change by 50 degrees F (Note: The Viscosity Regulator must be in limit while checking this).	Defective Viscosity Regulator or Viscosity Set-up Card. Replace the cards. Verify that the signal is wired to pin 48 on the temperature regulators and is not grounded.

TABLE 8.2 (Continued)
TROUBLESHOOTING CHART

SYMPTOMS	PROBABLE CAUSE	REMEDY
	ΔBP signal does not change machine Back Pressure. Verify this by rotating the ΔBP pot on the Viscosity Regulator during the screw rotate operation while observing the Pressure Guage on the machine machine. (Note: The Viscosity Regulator must be in limit while checking this, and the Back Pressure should change by approximately 50 PSI).	Back Pressure Setpoint is at minimum. Increase to a value that will allow the process control to have decrease correctional ability. Verify that the offset light on the Pressure Hydraulic Regulator comes on during the screw rotate operation to gate in the Δ BP signal. Verify the wiring of the Δ BP signal from the Viscosity Regulator Card to the Pressure Hydraulic Regulator Card.
	Δ Injection Rate signal does not change machine injection speed.	Verify that the offset light on the Flow Hydraulic Regulator comes on during the injection period to gate in the ΔIR signal. Verify that the Injection Rate Setting is not calling for more than the machine capability. (see instruction on Flow Control Function for scaling procedure). Verify that the Inject Pressure Setting is not so low that the Flow Control is pressure limited. Raise the Injection Pressure setpoint. The maximum Injection Rate correction is 20 percent of Injection Rate Setpoint therefore, verify that the Injection Rate Reference is being switched in to pin 44 of the Viscosity Regulator Card. Defective Viscosity Set-Up or Viscosity Regulator Card. Verify the wiring of the ΔIR signal from the Viscosity Regulator to the Flow Hydraulic Regulator.
VISCOSITY REGULATOR Limit Light Comes On	Power was applied to the panel with the Viscosity Regulator turned on.	Reset by operating the Regulator off-on switch to "off" and back to "on."
	A trend has set-up in the machine or process and the Regulator has been forced to consistently make corrections in the same direction.	For a New Set-Up Reset the Viscosity Regulator, let the zone heaters stabliize, and re-set up the Viscosity Regulator. For a Proven Set-Up Examine the machine with regard to a zone temperature out of range, or a hot runner heater or mold heater or cooler out of range. Verify that injection and holding pressures or speeds are not gradually changing.

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